

King Faisal University
College of Computer Sciences and Information Technology
Computer Science Department

[CS-210]: [Programming Methodologies]
Section: []

Lab [03]: [Selection: If Elese Statement]

Date: []

Answer Sheet

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Computer Number:

Part 1 — Guided Debugging & Code Analysis

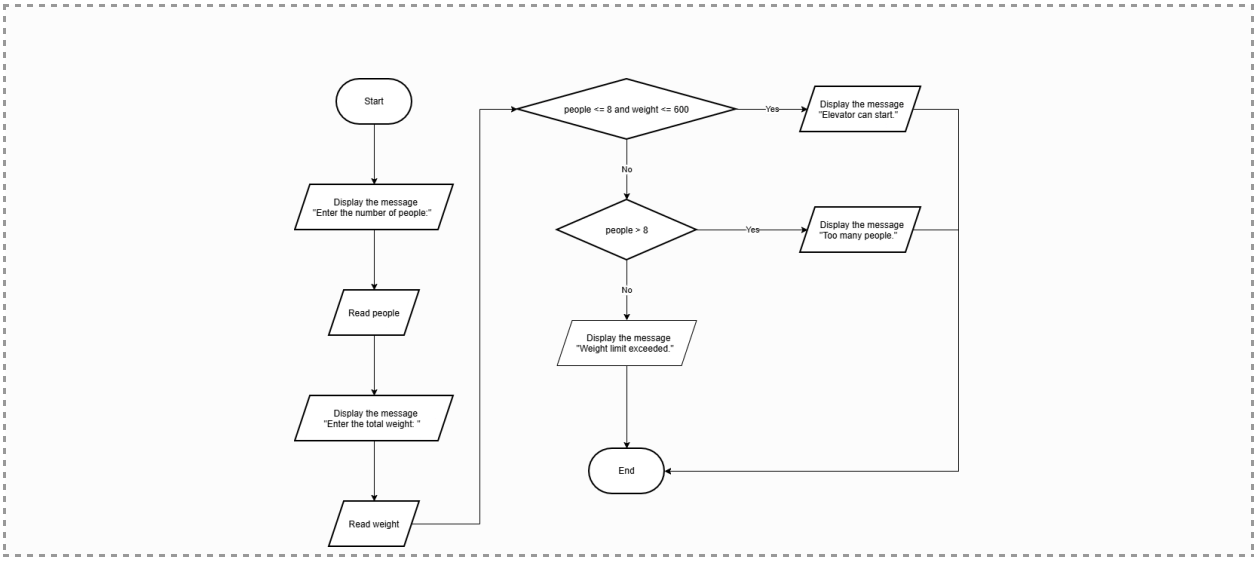
Error Identification & Correction Table

Fill in the table below by identifying the errors present in the code. One row is dedicated to each error.

Line #	Code Snippet with Error	Error Description (Student Fill)	Error Type (Syntax/Logical)	Corrected Code (Student Fill)
16	if people < 8 && weight < 600 {	No parenthesis for the condition.	Syntax	if (people < 8 && weight < 600) {
16	if (people < 8 && weight < 600) {	The condition doesn't accept 8 people nor weights of 600 KG, only below them.	Logical	if (people <= 8 && weight <= 600) {
19	else (people > 8 || weight > 600) {	The statement is not written correctly. Should be "else if".	Syntax	else if (people > 8 || weight > 600) {
19	else if (people > 8 || weight > 600) {	The condition doesn't express the output.	Logical	else if (people > 8) {

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



Corrected Source Code

Paste your complete, working, and corrected C++ source code in the box below.

```cpp
// Elevator Problem

#include <iostream>
using namespace std;

int main() {
    int people;
    double weight;

    cout << "Enter the number of people: ";
    cin >> people;

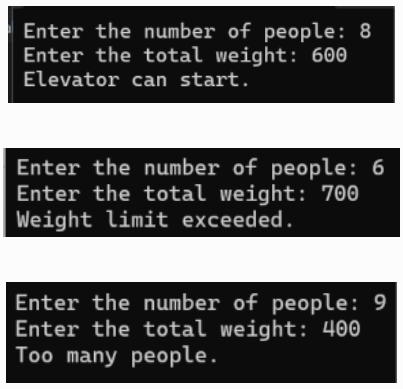
    cout << "Enter the total weight: ";
    cin >> weight;

    if (people <= 8 && weight <= 600) {
        cout << "Elevator can start." << endl;
    }
    else if (people > 8) {
        cout << "Too many people." << endl;
    }
    else {
        cout << "Weight limit exceeded." << endl;
    }

    return 0;
}
```

Execution Screenshot

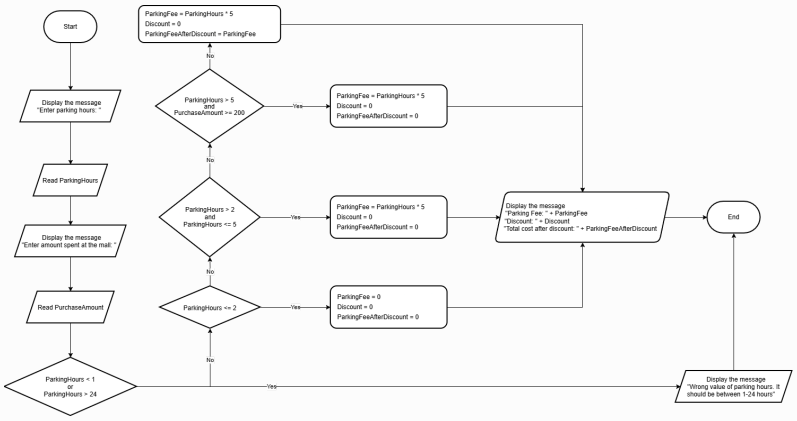
Paste a screenshot of your working program output here showing a successful temperature conversion.



Part 2 — Independent Program Implementation

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



C++ Source Code:

Paste your complete C++ source code for the VAT calculator in the box below.

```
1. #include <iostream>
2. #include <iomanip>
3. using namespace std;
4.
5. int main()
6. {
7.     unsigned short int ParkingHours;
8.     float PurchaseAmount;
9.
10.    cout << "Enter parking hours: ";
11.    cin >> ParkingHours;
12.
13.    cout << "Enter amount spent at the mall: ";
14.    cin >> PurchaseAmount;
15.
16.    if (ParkingHours < 1 || ParkingHours > 24)
17.    {
18.        cout << endl;
19.        cout << "Wrong value of parking hours." << endl;
20.        cout << "It should be between 1-24 hours" << endl;
21.    }
```

```
22.   else
23.   {
24.
25.     float ParkingFee, Discount, ParkingFeeAfterDiscount;
26.
27.     if (ParkingHours <= 2)
28.     {
29.       ParkingFee = 0;
30.        Discount = 0;
31.        ParkingFeeAfterDiscount = 0;
32.     }
33.     else if (ParkingHours > 2 && ParkingHours <= 5)
34.     {
35.        ParkingFee = ParkingHours * 5;
36.        Discount = 0;
37.        ParkingFeeAfterDiscount = 0;
38.     }
39.     else if (ParkingHours > 5 && PurchaseAmount >= 200)
40.     {
41.        ParkingFee = ParkingHours * 5;
42.        Discount = 0.15 * ParkingFee;
43.        ParkingFeeAfterDiscount = ParkingFee - Discount;
44.     }
45.     else // more than 5 hours of parking, but without paying 200 SAR.
46.     {
47.        ParkingFee = ParkingHours * 5;
48.        Discount = 0;
49.        ParkingFeeAfterDiscount = ParkingFee;
50.     }
51.
52.     cout << endl;
53.     cout << setprecision(2) << fixed; // to set the precision to 2 places.
54.     cout << "Parking Fee: " << ParkingFee << " SAR\n";
55.     cout << "Discount: " << Discount << " SAR\n";
56.     cout << "Total cost after discount: " << ParkingFeeAfterDiscount << " SAR\n";
57.   }
58.   return 0;
59. }
```

Execution Screenshot:

Paste a screenshot of your compiled VAT calculator running successfully with test data.

```
Enter parking hours: 2
Enter amount spent at the mall: 300

Parking Fee: 0.00 SAR
Discount: 0.00 SAR
Total cost after discount: 0.00 SAR
```

```
Enter parking hours: 5
Enter amount spent at the mall: 240

Parking Fee: 25.00 SAR
Discount: 0.00 SAR
Total cost after discount: 0.00 SAR
```

```
Enter parking hours: 6
Enter amount spent at the mall: 400

Parking Fee: 30.00 SAR
Discount: 4.50 SAR
Total cost after discount: 25.50 SAR
```

```
Enter parking hours: 6
Enter amount spent at the mall: 95

Parking Fee: 30.00 SAR
Discount: 0.00 SAR
Total cost after discount: 30.00 SAR
```